

PROJECT REPORT

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PROJECT REPORT

(Project Term January-May 2026)

(TrackIt – Study Session Monitoring System)

Submitted by

(Rahul Pandey)

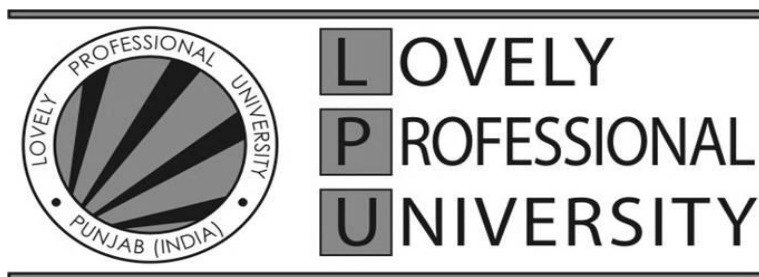
Registration Number:12316599

Course Code: INT 332

Under the Guidance of

(Dr. Alok Mishra)

School of Computer Science and Engineering



DECLARATION

I hereby declare that the project work entitled TrackIt – Study Session Monitoring System is an authentic record of my own work carried out as requirements of Project for the award of B.Tech Degree in Computer Science and Engineering from Lovely Professional University, Phagwara, under the guidance Dr. Alok Mishra, during January to May 2026. All the information furnished in this project report is based on my own intensive work and is genuine.

Name of Student: Rahul Pandey

Registration Number:12316599

(Rahul Pandey)
Date:29-05-2026

CERTIFICATE

This is to certify that the declaration statement made by the student is correct to the best of my knowledge and belief. He has completed this project under my guidance and supervision. The present work is the result of his original investigation, effort, and study. No part of this work has ever been submitted for any other degree at any University.

The project entitled "**TrackIt – Study Session Monitoring System**" is fit for submission and partial fulfilment of the requirements for the award of B.Tech Degree in Computer Science and Engineering from Lovely Professional University, Phagwara.

Dr. Alok Mishra

Designation

School of Computer Science and Engineering,
Lovely Professional University,
Phagwara, Punjab.

Date :29-05-2026

ACKNOWLEDGEMENT

I express my sincere gratitude to my project mentor for providing valuable guidance, support, and encouragement throughout the development of this project. Their suggestions and technical expertise greatly contributed to the successful completion of this work.

I also thank the School of Computer Science and Engineering, Lovely Professional University, for providing the necessary facilities and learning environment. Finally, I would like to thank my friends and family members for their continuous motivation and support.

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1. INTRODUCTION

TrackIt is a smart study session monitoring system developed to improve academic productivity and learning discipline among students. The application enables students to record and monitor their study activities while allowing teachers, mentors, and parents to track progress remotely.

The system is developed using Flutter, Firebase, Jenkins, Docker, NGINX, Prometheus, and Grafana technologies. Jenkins automates the CI/CD pipeline, Docker provides containerized deployment, NGINX hosts the web application, while Prometheus and Grafana provide monitoring and visualization of application and infrastructure performance.

Objectives

- a) Implement CI/CD using Jenkins.*
- b) Automate deployment using Docker containers.*
- c) Monitor application and server performance using Prometheus and Grafana.*
- d) Improve system reliability and scalability through DevOps practices.*

2 SCOPE OF THE STUDY (PROBLEM STATEMENT)

Students often struggle with maintaining study consistency, time management, and self-discipline. Traditional methods such as notebooks and manual records are inefficient and difficult to analyse.

The project aims to solve these issues by providing:

- Digital study session tracking.
- Real-time monitoring.
- Progress analysis through charts.
- Cloud-based data management.
- Performance reporting for students and viewers.

3 EXISTING SYSTEM

3.1 Introduction

Current study tracking methods mainly rely on manual record-keeping and self-reporting, which are time-consuming and inaccurate.

3.2 Existing Software

- Study planners
- Manual notebooks
- Basic timer applications
- Spreadsheet-based tracking systems

3.3 DFD for Present System

Current Flow:

Student → Manual Record → Teacher/Parent Review

Limitations:

- No real-time updates
- No automated analytics
- Difficult progress monitoring
- High chances of inaccurate data

3.4 What's New in the Proposed System

- Real-time monitoring
- Secure cloud storage
- Graphical analytics
- Student-viewer connectivity
- Performance tracking dashboard

4. PROBLEM ANALYSIS

4.1 Product Definition

TrackIt is a study monitoring platform that records study sessions, stores academic activity data, and provides real-time analytics to students and viewers.

4.2 Feasibility Analysis

Technical Feasibility

The project uses Flutter, Firebase Authentication, Firebase Realtime Database, Jenkins, Docker, NGINX, Prometheus, Grafana, cAdvisor, and Node Exporter. These technologies are widely adopted in modern software development and DevOps environments.

Economic Feasibility

Firebase provides affordable cloud services and reduces infrastructure costs.

Operational Feasibility

The application is user-friendly and requires minimal training.

4.3 Project Plan

Phase 1: Requirement Gathering

Phase 2: System Design

Phase 3: Frontend Development

Phase 4: Backend Integration

Phase 5: Testing

Phase 6: Deployment

5. SOFTWARE REQUIREMENT ANALYSIS

5.1 Introduction

Software Requirement Analysis identifies functional and non-functional requirements of the TrackIt system.

5.2 General Description

The system consists of:

- Student Module
- Viewer Module
- Authentication Module
- Analytics Module

5.3 Specific Requirements

Functional Requirements

- User Registration

- User Login
- Session Recording
- Subject Management
- Analytics Generation
- Viewer Monitoring

Non-Functional Requirements

- Security
- Reliability
- Scalability
- Performance
- Availability

Software Requirements

- Flutter SDK
- Dart
- Firebase
- Android Studio / VS Code
- Git & GitHub
- Docker
- Nginx

Additional DevOps Requirements

- Jenkins Server
- Prometheus Monitoring Server
- Grafana Dashboard Server
- cAdvisor
- Node Exporter

6. DESIGN

6.1 System Design

Developer

|

v

GitHub Repository

|

v

Jenkins Pipeline

|

v

Build Flutter Application

|

v

Docker Image Creation

|

v

NGINX Deployment

|

v

TrackIt Application

Monitoring Layer

Prometheus

|

v

Grafana

Metrics Sources:

- Node Exporter
- cAdvisor
- Docker Containers
- NGINX

6.2 Design Notations

Use Case Diagram

Actors:

- Student
- Viewer

Student Functions:

- Login
- Create Sessions
- View Analytics

Viewer Functions:

- Search Student
- Monitor Progress
- View Reports

6.3 Detailed Design

Student Dashboard

- Add Study Session
- Select Subject
- Track Duration
- View Charts

Viewer Dashboard

- Search Student
- Connect with Student
- Monitor Progress

6.4 Flowchart

Start



Login/Register



Student Dashboard



Create Study Session



Store Data in Firebase



Generate Analytics



Display Reports



Logout



End

6.5 Pseudo Code

BEGIN

User Login

IF Authentication Successful

Open Dashboard

Create Study Session

Store Session Data

Generate Analytics

Display Reports

ELSE

Show Error

END IF

END

6.6 Jenkins CI/CD Workflow

Jenkins Pipeline

The TrackIt project follows a Continuous Integration and Continuous Deployment (CI/CD) workflow using Jenkins.

Pipeline Steps:

1. Developer pushes source code to GitHub.
2. Jenkins automatically detects changes.
3. Jenkins builds the Flutter application.
4. Docker image is created.
5. Docker container is deployed.
6. NGINX serves the application.
7. Prometheus monitors system health.
8. Grafana displays performance dashboards.

Benefits:

- Automated deployment
- Reduced human error
- Faster software delivery
- Continuous monitoring

7. TESTING

7.1 Functional Testing

- Login Validation
- Session Creation
- Data Synchronization
- Analytics Generation

7.2 Structural Testing

- Code Review
- Module Testing

7.3 CI/CD Pipeline Testing

The Jenkins pipeline was tested to ensure:

- Successful code integration
- Automated build execution
- Docker image creation
- Automated deployment
- Monitoring service availability

7.4 Levels of Testing

Unit Testing

Testing individual modules.

Integration Testing

Testing interaction between Flutter and Firebase.

System Testing

Testing complete application functionality.

7.4 Testing Results

All modules functioned successfully without major errors.

8. IMPLEMENTATION

8.1 Implementation

Frontend Layer

- Flutter Web
- Dart

Backend Layer

- Firebase Authentication
- Firebase Realtime Database

CI/CD Layer

Jenkins

Jenkins automates:

- Source code integration
- Build process
- Docker image generation
- Application deployment

Containerization Layer

Docker

Docker packages the application into portable containers for consistent deployment.

Hosting Layer

NGINX

NGINX serves the Flutter Web application and handles incoming requests efficiently.

Monitoring Layer

Prometheus

Prometheus collects:

- CPU metrics
- Memory metrics
- Container metrics
- Server health metrics

Grafana

Grafana visualizes:

- Real-time dashboards
- Infrastructure performance
- Application monitoring reports

9. PROJECT LEGACY

9.1 Current Status

The TrackIt platform has been successfully implemented with a complete DevOps workflow. Jenkins automates the CI/CD process, Docker enables containerized deployment, NGINX hosts the application, and Prometheus-Grafana provide monitoring and observability.

9.2 Remaining Areas of Concern

- AI-based recommendations not yet implemented.
- Notification system can be enhanced.
- Attendance tracking can be integrated.

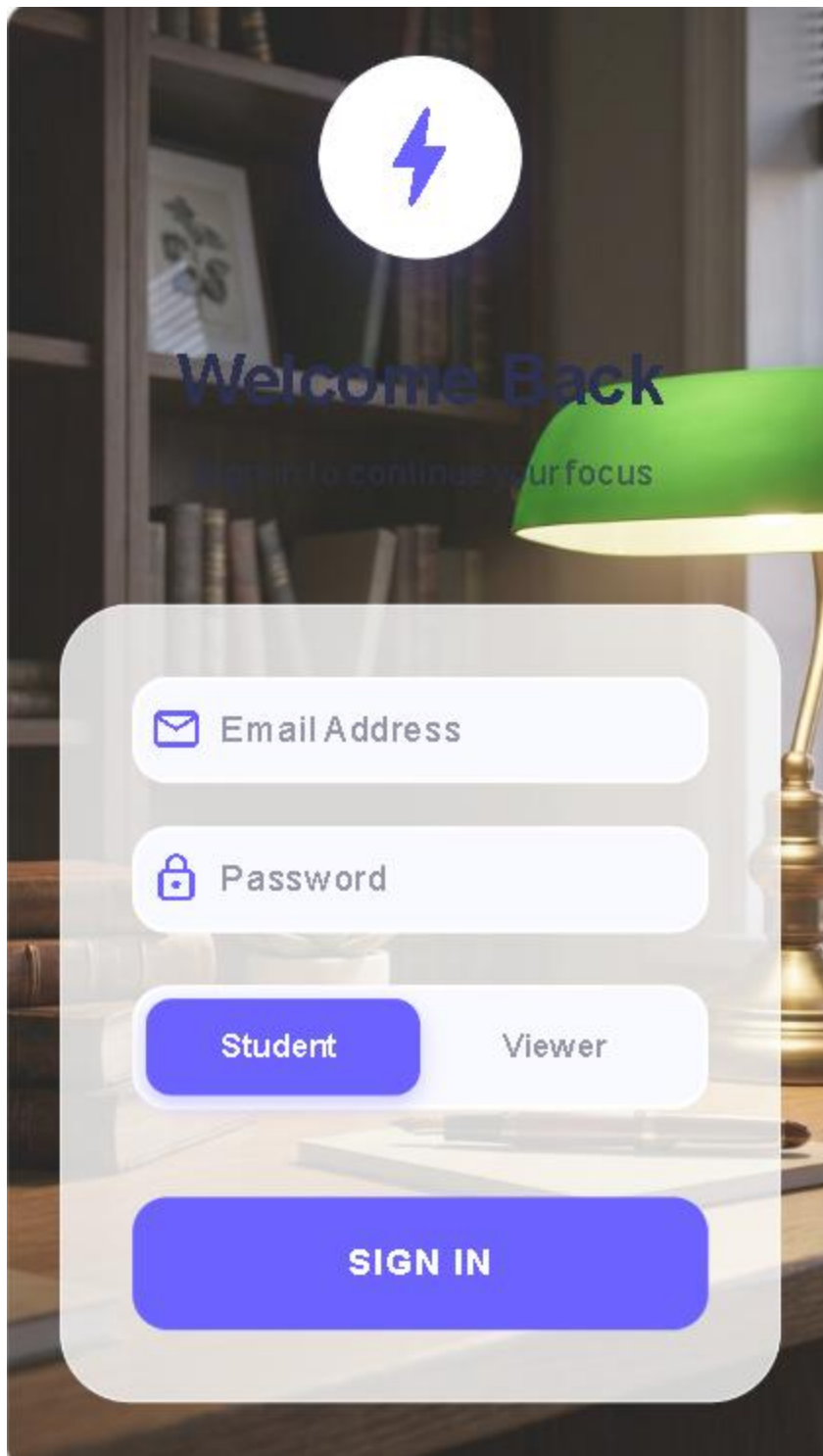
9.3 Technical and Managerial Lessons Learned

- Flutter cross-platform development.
 - Firebase cloud integration.
 - Real-time database management.
 - Team collaboration and project planning.
-

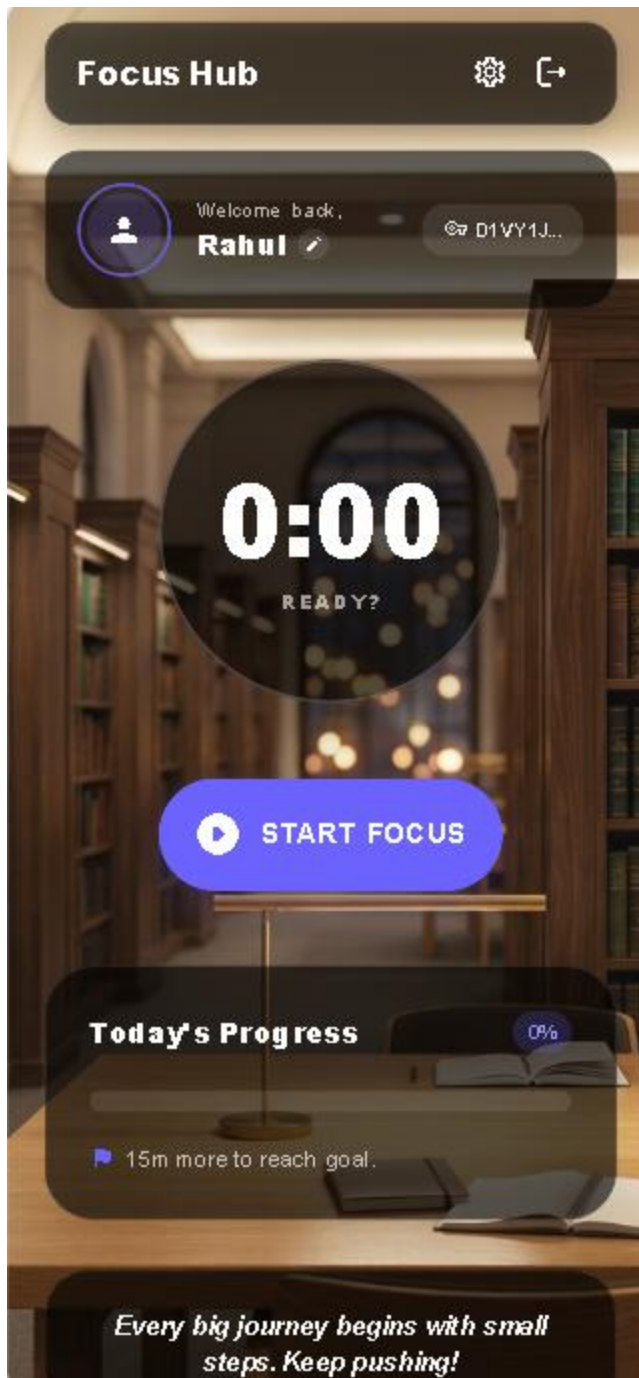
10. SYSTEM SNAPSHOTS

Insert screenshots of:

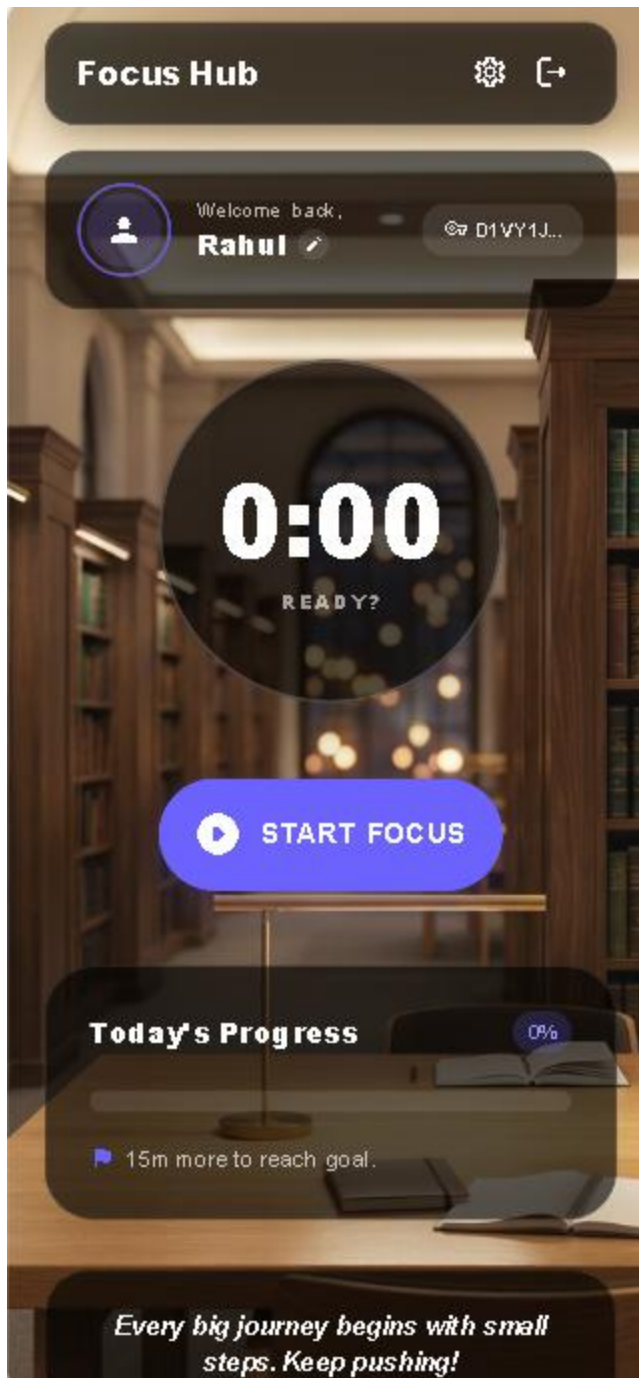
1. Login Screen



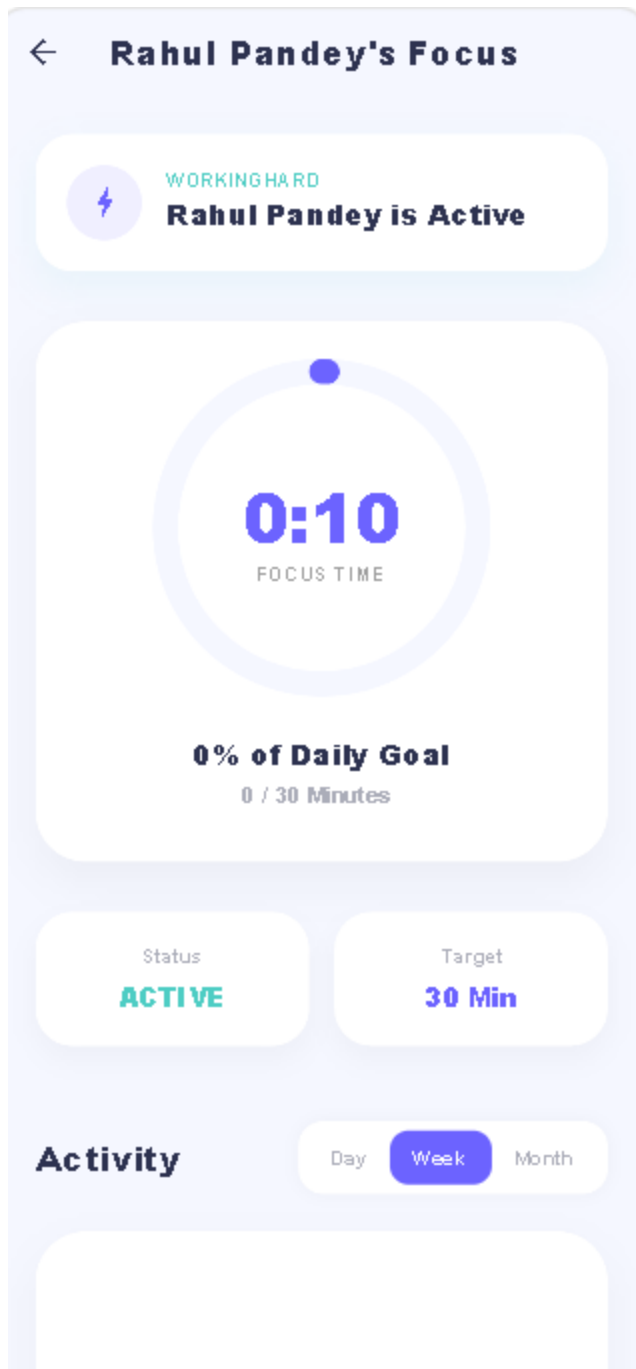
2. Registration Screen



3. Student Dashboard



4. Viewer Dashboard



5. Jenkins Pipeline

The screenshot shows the Jenkins web interface for job #17. The job is in a successful state, indicated by a green checkmark. The build was started by an SCM change and took 8 minutes and 10 seconds to complete. The console output shows the build process, and the pipeline steps are visible at the bottom. The interface includes a sidebar with navigation options like Status, Changes, Console Output, and Pipeline Overview. The main content area displays the build details, including the revision and repository information.

Jenkins / TrackIt-Deploy / #17

Status ✔ **#17 (25 May 2026, 08:46:07)** Add description Keep this build forever

Changes Started by an SCM change Started 4 days 0 hr ago Took 8 min 10 sec

Console Output

Edit Build Information

Delete build '#17'

Polling Log

Timings

Git Build Data

Pipeline Overview

Replay

Pipeline Steps

git **Revision:** 3216b65bce5f06a7503482cb6162f61df397cd5 **Repository:** <https://github.com/raahul-pandey-ce/TrackIt.git>

refs/remotes/origin/main

Title Updated 3216b65 **Rahul Pandey** at 08:45 25/05/2026

6. Grafana

The screenshot shows the Grafana web interface displaying the Node Exporter Full dashboard. The dashboard is configured to show metrics for the 'node-exporter' job. The 'Quick CPU / Mem / Disk' section displays several gauges for CPU usage, memory usage, and disk usage. The 'Basic CPU / Mem / Net / Disk' section displays two line graphs for CPU and memory usage over time. The interface includes a sidebar with navigation options like Home, Bookmarks, Starred, and Dashboards. The main content area displays the dashboard metrics and graphs.

Grafana **Dashboards** > Node Exporter Full

Datasource prometheus **Job** node-exporter **Nodename** 2aaa5cb52d57 **Instance** node-exporter:9100

Quick CPU / Mem / Disk

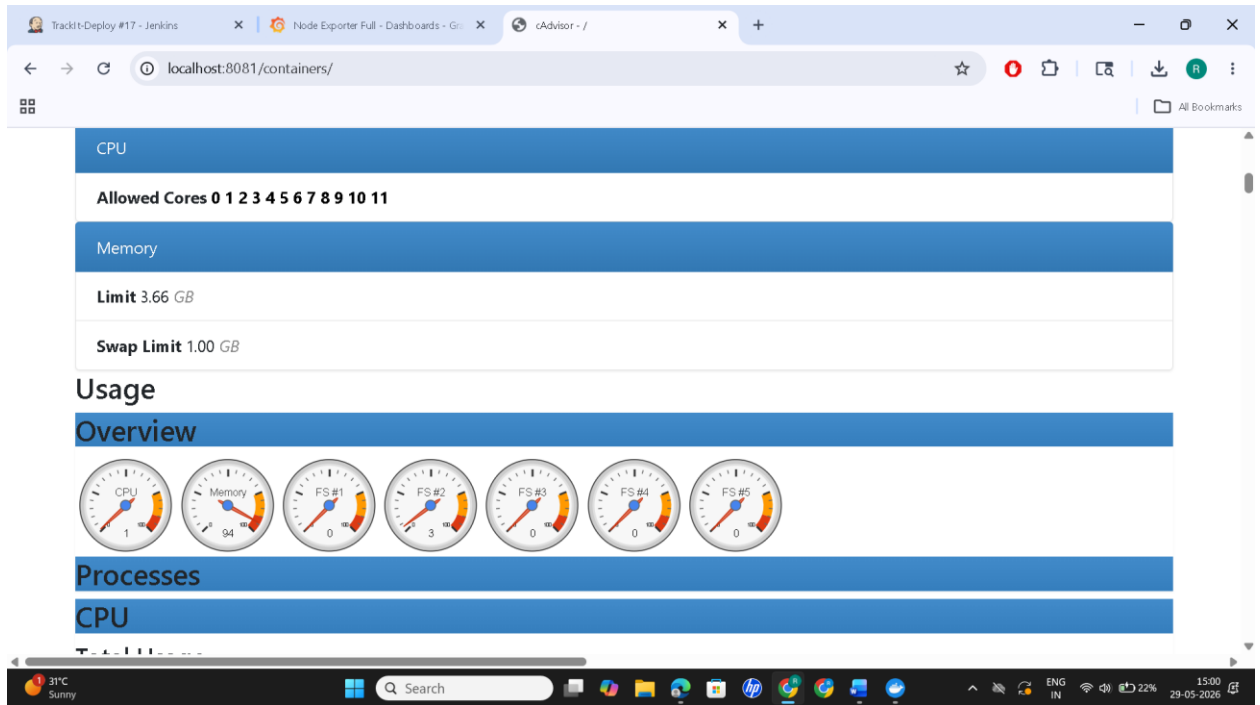
Metric	Value
CPU	0.2%
Mem	0.0%
Disk	0.0%
Uptime	6.2 mins

Basic CPU / Mem / Net / Disk

CPU Basic

Memory Basic

7. Monitoring



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